Definitional Instability in Kinase Inhibitor Selectivity: An Empirical Characterization and Motivated Properties for a Well-Formed Measure

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Abstract

Kinase inhibitor selectivity is quantified using several competing definitions, including the S-score, selectivity entropy, Gini coefficient, and ratio-based measures, which are often used interchangeably. We propose four properties that a well-formed selectivity measure should satisfy: a reliability threshold, bounded sensitivity to the reference off-target, consistency across placements of the activity boundary, and monotonicity when a weakly bound kinase is added. None of the four definitions satisfies all of them. The properties are motivated by four profiling datasets spanning three assay technologies (68–704 compounds each), on which rank instability is associated with binding profile shape and has three sources: conflicting scores for compounds with no binding above the activity threshold, near-tied top targets, and distribution-ratio disagreement for promiscuous compounds. With entropy and Gini on a common log-affinity scale, the S-score, entropy and Gini agree more with one another than with the fold-selectivity ratio, a grouping that fails for the entropy as originally published. A gated, smooth-hinge variant of the entropy satisfies all four properties when its activity

boundary is fixed at the assay floor, ranks these datasets almost identically to the entropy, and recovers its full-panel ranking from about a tenth to a quarter of a panel on three of the four datasets.

Keywords: binding selectivity, cancer drug discovery, cheminformatics, kinase inhibition, research methodology.

Introduction

Kinase inhibitors are among the most intensely investigated compound classes in drug discovery, with 80 FDA-approved agents and several hundred in clinical development.[1] Because the human kinome is comprised of more than 500 proteins sharing a highly conserved ATP-binding site,[2] the design of inhibitors with controlled target profiles remains a central challenge. Selectivity is a primary criterion in lead optimization and a key determinant in the drug approval process.[3]

Despite its importance, no single consistent definition of selectivity exists. The field employs at least four distinct quantitative measures: the S-score,[4] selectivity entropy,[5] Gini coefficient,[6] and ratio-based measures, which compare the primary target with an off-target. These definitions are often used interchangeably, and their parameter values (the activity threshold in the S-score, the baseline in the entropy and Gini coefficient, the reference off-target in ratio-based measures) are usually set by convention. Standardizing them would remove one source of disagreement between laboratories and platforms.

Bosc et al.[7] showed that these definitions can produce substantially different compound rankings on the same dataset, and the KInhibition platform[8] explicitly acknowledged that conflicting results across metrics are common. To our knowledge, which binding profile properties determine when the definitions agree or conflict, and why, has not been characterized. Go/no-go decisions in lead optimization are often made on selectivity assessments whose sensitivity to the choice of definition is unknown, and inconsistent definitions make

compounds hard to compare across studies.

In information theory, Shannon[9] settled which function should measure entropy by writing down a few properties a sensible measure must have and showing that only one formula satisfies them. We follow this precedent in part: we characterize how the competing measures behave, state properties a good measure should have, and propose a formula that satisfies them. We use four profiling datasets that span three assay technologies. Our contributions are as follows:

(i) we show that, with entropy and Gini on a common log-affinity scale, the S-score, entropy and Gini (the distribution-based family) agree more with each other on all four datasets (Spearman 0.72–1.00) than with the fold-selectivity ratio $R(k)$ (0.14–0.62). With the entropy in its original association-constant form, this grouping fails on all three affinity datasets (Methods);

(ii) we identify three sources of rank instability: no binding above the activity threshold, near-tied top targets, and broad moderate off-target activity, often with one dominant target;

(iii) we motivate four properties, (D1) a reliability threshold, (D2) bounded gap sensitivity, (D3) distributional consistency and (D4) monotonicity under weak off-target addition, and find that none of the four definitions assessed satisfies all of them. D1, a reporting convention that any of them could adopt, is the only one they all violate;

(iv) we propose a candidate measure, a gated, smooth-hinge effective-target number, which shows that the four properties can be met together (D3 only with its activity boundary fixed at the assay floor). It ranks these datasets almost identically to the entropy but, unlike it, provably cannot be made to look more selective by adding a weakly bound kinase. It recovers its full-panel ranking from about a tenth to a quarter of the panel on three of the four datasets.

Background

Existing Selectivity Metrics and Their Comparison

Several metrics have been proposed to quantify kinase inhibitor selectivity from profiling panel data. Karaman et al.[4] introduced the selectivity score (S-score), defined as the fraction of tested kinases bound with a dissociation constant below a fixed cutoff. Uitdehaag and Zaman[5] proposed the selectivity entropy, which treats the normalized binding profile as a probability distribution and computes its Shannon entropy. Graczyk[6] adapted the Gini coefficient from economics, treating the distribution of a compound's activity across the panel as an inequality distribution. Klaeger et al.[10] introduced the concentration- and target-dependent selectivity (CATDS) score, which quantifies the reduction in binding to a given target relative to the summed reduction across all detected targets at a specified inhibitor concentration (see Discussion for why we did not implement it as a fifth measure).

Bosc et al.[7] subsequently proposed two further metrics, the window score and ranking score, and compared them against four previously published metrics (the S-score, Gini coefficient, selectivity entropy, and a partition index) on the Davis and Anastassiadis datasets, showing that compound rankings differ substantially depending on the metric applied and that certain compounds cannot be scored under some definitions when threshold conditions are not satisfied. For example, ratio-based scores are undefined for a compound with no kinase above the activity threshold, since there is then no primary target or off-target to compare. Our implementation of $R(k)$ does not follow this convention (Methods, "Ratio").

To our knowledge, attempts to reconcile the definitions have been limited. Bajorath et al.[11] compared CATDS-based selectivity assessments from Klaeger et al. with activity data mined from ChEMBL for the same compounds, finding broad consistency between the two sources. The KInhibition platform[8] addressed conflicting results across metrics by introducing its own composite score (combining on-target inhibition with an off-target inhibition penalty). More recently, the same group built a KIRHub resource[12] on a new profiling campaign of

86 approved kinase inhibitors against 758 kinases and disease-associated mutant variants. This resource offers users a choice between a further named metric (the KISS score, an on-target/off-target balance) and a separate Gini-based score.

Published comparisons evaluate each metric at fixed parameter values, so, to our knowledge, the sensitivity of rankings to parameter choice has not been characterized systematically. The present work addresses this, together with the question raised in the Introduction of which profile properties determine when the definitions agree.

Axiomatic Approaches to Ranking and Measurement

Precedent for an axiomatic characterization of measurement functions is established. Shannon[9] demonstrated that the entropy of a probability distribution is uniquely determined, up to a multiplicative constant, by three requirements: continuity in the probabilities, monotonic growth with the number of equally likely outcomes, and consistency under grouping of outcomes. Khinchin[13] subsequently gave an alternative axiomatization (continuity, maximality on the uniform distribution, expansibility, and additivity over independent subsystems). The selectivity entropy of Uitdehaag and Zaman applies the Shannon formula to normalized binding profiles but provides no analogous axiomatic justification. The axioms that single out Shannon entropy, such as maximality on the uniform distribution and additivity over independent subsystems, characterize a measure of uncertainty, and none of them was chosen with selectivity in mind.

Arrow[14] established that no social welfare function satisfies simultaneously the requirements of unrestricted domain, Pareto efficiency, independence of irrelevant alternatives, and non-dictatorship. The theorem is relevant here in two respects: it shows that axiomatic analysis of a ranking concept can yield impossibility, and that competing ranking functions can reflect trade-offs between properties that cannot all be met at once. Singleton and Booth[15] applied this methodology to truth discovery, i.e. the problem of identifying true facts from conflicting reports of unknown reliability, posing it as a social choice problem,

formulating candidate axioms, and proving that a subset cannot be jointly satisfied.

To our knowledge, this axiomatic style of analysis has not previously been applied to any selectivity or promiscuity measure in drug discovery. The present work proposes empirically motivated properties as a foundation for such an analysis.

Methods

Datasets

Our primary analysis uses two publicly available kinase inhibitor profiling datasets, described first. The Davis dataset[16] comprises binding affinities (K_d) for 68 small-molecule kinase inhibitors measured against 433 recombinant human kinase constructs in a competition-binding assay. Affinity values are reported as dissociation constants and were converted to $\mathrm{p}K_d = -\log_{10}(K_d/10^9)$ prior to analysis, following the convention of the DeepDTA benchmark.[17] The resulting affinity matrix is complete (68×433, 29,444 entries), with all non-binding pairs assigned the detection-floor value $\mathrm{p}K_d = 5.0$ (corresponding to a K_d of 10 μM). This is the standard convention for the dataset.[7,17] It places non-binding pairs at the floor, above their true, unmeasured affinity.

Active entries (pKd > 6, i.e. $K_d < 1$ μM) make up 17.9% of the Davis matrix. This 1 μM cutoff is a common activity threshold in kinase profiling. A stricter cutoff ($\mathrm{p}K_d > 7$, $K_d < 100$ nM) shrinks per-compound active sets substantially (mean active kinases per compound falls from 77.5 to 34.4, a 2.25-fold reduction) and pushes two of the 68 Davis compounds, which have at least one active kinase at $\mathrm{p}K_d > 6$, to zero active kinases at the stricter cutoff (into the zero-active regime discussed in Results). Shifting the S-score's threshold sweep up by one log unit (from $[5.5, 8.0]$ to $[6.5, 9.0]$) to match this cutoff lowers the S-score's correlation with entropy and Gini from ≈ 0.95 to 0.83 and 0.84 and raises its correlation with the ratio from 0.456 to 0.574 (the entropy, Gini, and ratio sweeps are themselves unchanged by this check). The S-score therefore stays much closer to entropy and

Gini than to the ratio.

The Klaeger study[10] profiled 243 clinical kinase inhibitors against the human kinome using Kinobeads chemical proteomics. Its Kinobeads table lists fitted targets for 229 of them. Apparent dissociation constants (K_d^{app}) were derived from dose–response curves across eight inhibitor concentrations in cancer cell lysates. Only high-confidence measurements (as annotated by the original authors) were retained, yielding 5,123 drug–target pairs for 222 drugs and 343 kinase targets. Affinity values were converted to $\mathrm{p}K_d$ using the same formula as above. Missing entries, combinations for which no binding was detected, were assigned the background value $\mathrm{p}K_d = 5.0$. The resulting Klaeger matrix is 222×343. Active entries (pKd > 6) make up 3.6% of the Klaeger matrix (Davis: 17.9%). The 46 compounds profiled in both datasets (Discussion) are active against 16.5% of the Davis panel on average and 3.2% of the Klaeger panel, so the difference comes from the assay. These data cannot tell whether it comes from the chemoproteomic assay's detection threshold or from which kinases the assay captures.

The high-confidence filter removes 793 of the 5,916 drug–target rows in the table (13.4%), one of which has no numeric K_d^{app}. A removed pair is treated identically to one with no detected binding and assigned the same $\mathrm{p}K_d = 5.0$ background value. Of the 792 low-confidence pairs with a numeric K_d^{app}, 561 fall within the 222×343 high-confidence matrix. The other 231 involve 7 compounds and 177 kinases with no high-confidence measurement. Adding the 561 pairs to the same matrix changes only 0.74% of its entries, shifts every cell of Table 1's Klaeger triangle by at most 0.05, and leaves every Klaeger panel-size p^* value (Results) unchanged. Retaining all 792 pairs also adds those compounds and kinases, which changes the panel itself. It shifts the Klaeger cells of Table 1 by up to 0.12 (S-score against ratio, from 0.22 to 0.34) and leaves the two-family structure intact. The filter choice is therefore not responsible for the reported correlations or convergence thresholds.

To test whether our findings depend on the assay technology, we added two further datasets that measure kinase inhibition by different means. The Anastassiadis dataset[18]

profiles 178 inhibitors against 300 kinases in a single-concentration (0.5 μM) radiometric activity assay, reported as percent residual kinase activity relative to a solvent control. We converted these to percent inhibition (100 minus residual activity, clipped to $[0, 100]$, where higher values denote stronger inhibition), with the 1.1% missing entries set to zero inhibition. The Metz dataset[19] reports pK_i affinities for a large medicinal-chemistry campaign using the same negative-log scale as pK_d. The value p$K_i = 4.0$ is the lowest value reported anywhere in the released data (only 0.17% of measurements), so we assign it to unmeasured combinations as the assay's own detection floor (the same convention used for the Davis p$K_d = 5.0$ floor above). Coverage per compound is highly skewed (a median compound is measured against only 21 of the 172 kinases). Because of this, we restricted the dataset to the 704 compounds profiled against at least 50 of the 172 kinases ($\geq$ 29% coverage). The threshold excludes the most sparsely profiled compounds while retaining a large enough compound set for the panel-size and instability analyses. Across the 704 retained compounds, coverage ranges from 29.1% to 83.1% of the 172 kinases (mean 40.9%, median 37.2%, i.e. 64 of 172 kinases). Thus, for the median retained compound, most entries are floor-imputed, and fewer than a quarter of compounds have more than half of their entries measured (75th percentile 47.1%). A sizable minority sit close to the threshold: 23 of 704 (3.3%) have exactly 50 measured kinases, and 170 (24.1%) have 55 or fewer.

Floor imputation has a large effect on Metz. Nearly every measured value (99.9%) lies above the floor, so a compound measured against more kinases has more kinases with positive weight and looks less selective. Across the 704 compounds, the Spearman correlation between a compound's coverage and its median rank (rank 1 = most selective) is 0.98 for the entropy, 0.95 for the Gini, 0.76 for the S-score, and 0.11 for the ratio. Part of this is real, since compounds measured against more kinases are also active against a larger share of the kinases actually measured, but that association is much weaker ($r = 0.33$). No fully measured subset of useful size exists (only 7 kinases are measured for at least 80% of the compounds), so we keep Metz as a fourth dataset but read its distribution-based results, and their low agreement

with the ratio, as largely reflecting coverage.

The four datasets span three assay technologies (competition binding, chemoproteomics, and functional activity) and two readout types (affinity and single-concentration inhibition). Each selectivity definition is computed on the native readout of its dataset, and every comparison between datasets uses rank correlations computed within each dataset. For the percent-inhibition and pK_i datasets, the S-score threshold, and entropy/Gini baseline grids were shifted to span each dataset's active range (entropy/Gini baselines over $[0, 30]\%$ inhibition on Anastassiadis and p$K_i \in [4.0, 5.5]$ on Metz, against $[5.0, 6.5]$ on the pK_d datasets), while the ratio off-target index $k \in \{1, \ldots, 5\}$ was left unchanged (it is an index of off-targets and so needs no rescaling).

We excluded two additional candidate datasets. The Gao screen[20] is a single-concentration functional inhibition assay of the same readout type as the Anastassiadis dataset already included, and its raw data are not openly redistributable. Including it would add a second instance of an assay type rather than a new assay technology. The Patricelli study[21] profiles only a small number of inhibitors against native kinases as a proof of concept for the KiNativ platform. A ranking-stability analysis requires many compounds to produce a meaningful ranking, so this dataset is not suited to the present comparison.

Selectivity Definitions and Parameterization

We implemented four selectivity definitions, each with parameters whose values are not standardized in the literature. Throughout, a compound's binding profile is the vector of its affinities p$K_{d,i}$ across the N panel kinases, with higher pK_d denoting tighter binding.

S-score. Karaman et al.[4] defined the selectivity score as the fraction of the panel that a compound binds: the number of kinases bound with a dissociation constant below a cutoff, divided by the number of distinct kinases tested. They reported $S(3\ \mu\text{M})$ and $S(100\ \text{nM})$, for K_d cutoffs of 3 μM and 100 nM. We implemented the same fraction, with the cutoff expressed

as a threshold τ on pK_d:

$$S(\tau) = \frac{1}{N} \sum_{i=1}^{N} \mathbf{1}[\mathrm{p}K_{d,i} > \tau]. \tag{1}$$

Here N is the number of kinases in the panel. On Metz this includes kinases a compound was not tested against, which are imputed at the floor and so count only in the denominator. We used $\tau \in \{5.5, 5.75, \ldots, 8.0\}$ (11 values), spanning K_d cutoffs from approximately 3 μM to 10 nM, which includes both of Karaman et al.'s cutoffs ($\tau \approx 5.5$ and $\tau = 7$).

Selectivity entropy. Uitdehaag and Zaman[5] treat the compound's binding across the panel as a probability distribution and report its Shannon entropy. A profile concentrated on few kinases has low entropy and is selective. In the original definition, the weight on each kinase is its association constant $K_{a,i} = 1/K_{d,i}$ (the share of compound that would be bound to kinase i in an equimolar mixture), so $p_i = K_{a,i} / \sum_j K_{a,j}$. We implemented the entropy on the same log-affinity scale used by the other metrics, weighting each kinase by its affinity in excess of a baseline β:

$$H(\beta) = -\sum_{i=1}^{N} p_i \log_2 p_i, \quad p_i = \frac{\max(\mathrm{p}K_{d,i} - \beta,\, 0)}{\sum_j \max(\mathrm{p}K_{d,j} - \beta,\, 0)}, \tag{2}$$

where β is the affinity below which binding is treated as negligible. This makes β an explicit parameter. We swept $\beta \in \{5.0, 5.25, \ldots, 6.5\}$ (7 values), a narrower range than the S-score's τ sweep. Raising either parameter leaves a growing share of compounds with no kinase above it (on Klaeger, 7.2% of compounds at p$K_d = 6.0$ and 55% at p$K_d = 8.0$), where the S-score becomes 0 and the entropy and Gini become undefined. The τ range is wider because it was chosen to include Karaman et al.'s cutoffs. The baseline β has no published reference points and was kept near the p$K_d = 6$ activity threshold. We compare this weighting with the original association-constant weighting below.

Gini coefficient. Graczyk[6] adapted the Gini coefficient from economics: kinases are sorted by activity and the cumulative fraction of total binding is plotted against the cumulative

fraction of kinases (a Lorenz curve). The Gini coefficient, which measures how unequally binding is distributed across the kinome, is twice the area between this curve and the diagonal of equal distribution. The original work computes it from percent-inhibition values. As with the entropy, we computed it on the baseline-subtracted log-affinity profile, using the standard discrete formula:

$$G(\beta) = \frac{2\sum_{i=1}^{N} i \cdot x_{(i)}}{N\sum_{i=1}^{N} x_{(i)}} - \frac{N+1}{N}, \qquad x_{(i)} = \max(\mathrm{p}K_{d,i} - \beta,\, 0) \text{ sorted ascending.} \tag{3}$$

We swept β over the same range as for the entropy. A compound whose binding is concentrated on one kinase has a highly unequal distribution (G is near 1), while a compound that binds many kinases equally has G near 0. Higher values therefore indicate greater selectivity.

Ratio. Classical pharmacological selectivity is pairwise. It quantifies how much more tightly a compound binds its primary target than an off-target. Karaman et al.[4] also reported a score of this kind alongside the S-score: the fraction of tested kinases bound within tenfold of the affinity for the compound's intended primary target, which combines a ratio cutoff with an S-score-style count. Anchoring on an annotated primary target avoids the ratio's near-tie instability (Type 2, Results; D2, below). The score, however, has its own unstandardized parameter (the tenfold cutoff) and, as a fraction of the panel, violates D4 in the same way as the S-score. We instead implemented the pairwise comparison from the binding profile alone, as every measure compared here is a function of the profile (Motivated Properties, below). We included the reference off-target as an explicit parameter whose effect can be studied directly: the fold-selectivity, in log units, between the primary target and the k-th ranked off-target:

$$R(k) = \mathrm{p}K_{d,(1)} - \mathrm{p}K_{d,(k+1)} = \log_{10} \frac{K_{d,(k+1)}}{K_{d,(1)}}, \tag{4}$$

where $\mathrm{p}K_{d,(j)}$ is the j-th largest affinity in the profile and $k \in \{1, 2, 3, 4, 5\}$ selects the reference off-target. We capped k at 5 because, on the sparser Klaeger panel, a large share of

compounds already have fewer than five active kinases (median 5.5; 45% have fewer than five). For those compounds, a larger k would compare the primary target against the assay floor instead of an off-target.

Our $R(k)$ thus scores every compound, substituting the floor for a missing off-target. This departs from Bosc et al.'s convention (Background), so $R(k)$'s D1 violation comes from this implementation choice.

The concentration- and target-dependent selectivity (CATDS) score of Klaeger et al.[10] is not a ratio in this sense, since it normalizes over the whole panel (Background), and on our data it behaves like a member of the distribution-based family (Discussion, Supporting Information Table S2). We therefore represent the ratio family by $R(k)$ alone.

Faithfulness to the original definitions. The S-score and the ratio are computed in the form given by their original sources (a panel-hit fraction and a primary-to-off-target affinity ratio, respectively). For the entropy and the Gini coefficient, we used the baseline-subtracted log-affinity profile in place of the original association constants (entropy) and percent-inhibition values (Gini), so that all four metrics operate on one common scale with the baseline β as an explicit parameter. Every result reported for the entropy and the Gini coefficient elsewhere in this paper, including Table 1's two-family clustering, therefore concerns this baseline-subtracted form. To test whether this choice supports our conclusions, we recomputed the entropy and Gini with association-constant weighting, then compared the resulting rankings with the log-affinity forms. For the entropy, this is the original form, while for the Gini, it is an affinity-scale analogue. In Figure 1 the association-constant forms are computed over every entry of the matrix, floor-imputed ones included. The two operationalizations differ substantially: for entropy, Spearman $r = 0.42$ (Davis) and -0.34 (Klaeger), while for Gini, $r = 0.11$ and -0.45 (Figure 1).

As a further robustness check, we recomputed the full cross-definition correlation matrix of Table 1 with entropy replaced by its association-constant form on Davis, Klaeger, and

Metz. The Gini coefficient was replaced by its native percent-inhibition form (no baseline subtraction) on Anastassiadis, the one dataset where that native scale exists. On Klaeger and Metz, which have missing data, this association-constant form was computed per compound over only the pairs with a reported affinity, so that pairs without one (no binding detected on Klaeger; not tested on Metz) are excluded from the weighting. The Davis dataset is released as a complete matrix, with non-binding pairs already set to the floor, so every entry is used there.

On Davis, entropy decouples from the rest of the distribution-based family (S-score $r = 0.42$, Gini $r = 0.42$), close to its correlation with the ratio ($r = 0.34$), consistent with Figure 1. On Klaeger and Metz, entropy correlates more strongly with the ratio ($r = 0.79$ and 0.76) than with the S-score ($r = 0.57$ and 0.29) or the manuscript's own log-affinity Gini ($r = 0.69$ and 0.39). The native percent-inhibition Gini on Anastassiadis stays only partly consistent with the log-affinity form. It correlates with the entropy ($r = 0.63$), but weakly, and about equally with the S-score ($r = 0.35$) and the ratio ($r = 0.31$). So, the two-family pattern is not clearly preserved there either.

We draw two conclusions from this. First, the two-family structure we report throughout is a property of the operationalization we adopt. It does not hold for the entropy and Gini as originally published. Second, the direction of the effect is dataset-dependent, so no single correction recovers one stable cross-dataset answer. The operationalization choice is itself a source of definitional instability, in addition to the three we characterize in Results. Full correlation matrices for all four datasets are reported in Supporting Information (Table S3) and in the code repository.

Rank Stability Analysis

For each definition and each parameter value, compounds were ranked from most selective (rank 1) to least selective (rank n, where n is the number of compounds in the dataset). The S-score and the raw Shannon entropy both run opposite to this convention (a higher fraction

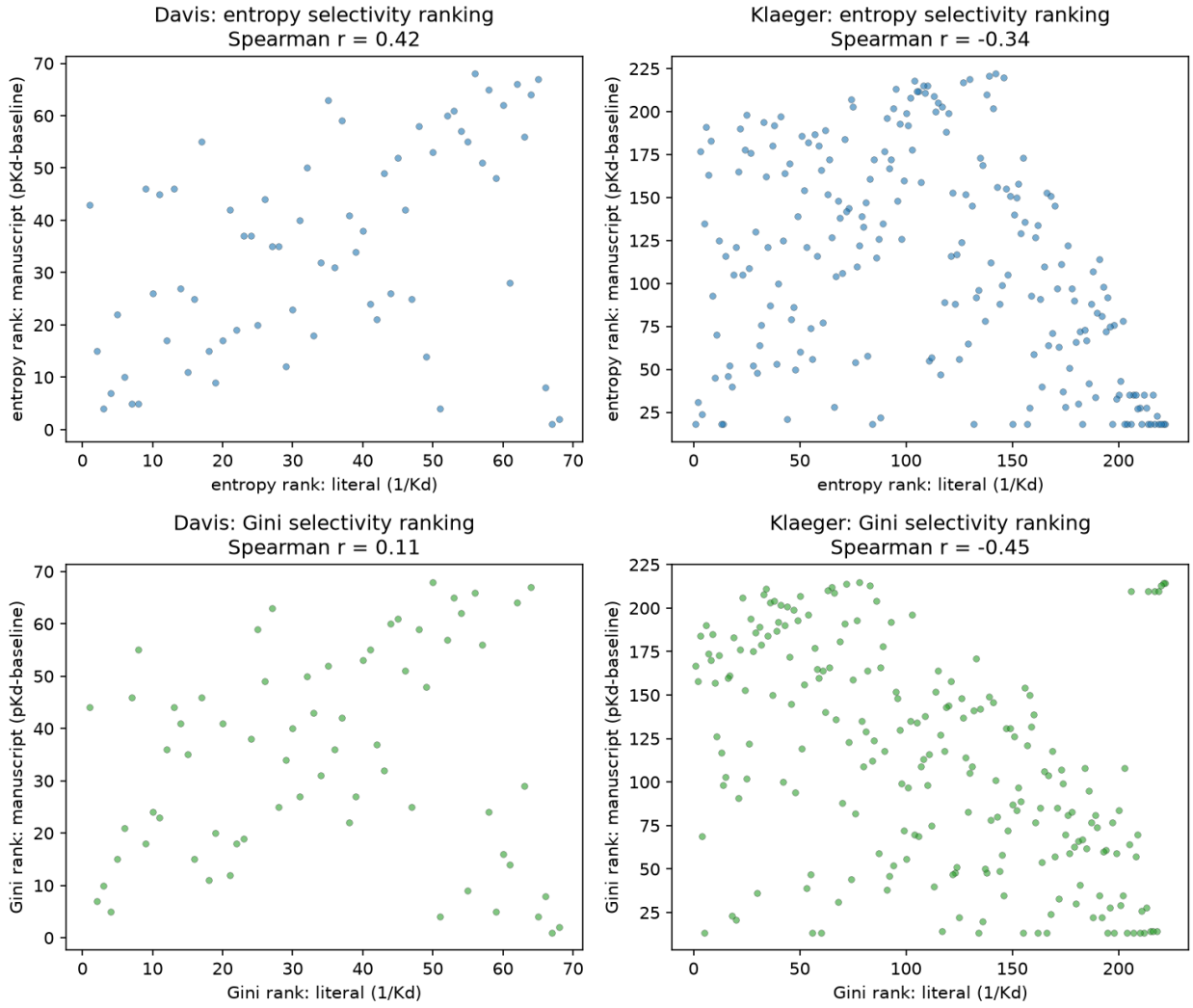


Figure 1: Entropy (top) and Gini coefficient (bottom) rankings on Davis and Klaeger under two operationalizations: a form close to the original publication (association-constant weighting, x-axis) and the baseline-subtracted log-affinity form used in the main text (y-axis). Spearman r above each panel.

of panel hits, or a higher raw entropy, indicates a *more* promiscuous compound), so both are negated before ranking throughout this paper. The Gini coefficient and $R(k)$ already increase with selectivity and are used unnegated. Ties were broken by average rank. Where the entropy and Gini are undefined (no kinase above the baseline β), our implementation sets both scores to 0, which ranks the compound as maximally selective under the (negated) entropy measure, tied with single-target profiles, and as minimally selective under the Gini. A compound can be undefined at some baselines of the sweep and defined at others. Results compares this with computing instability only over the configurations in which a compound's score is defined. The ratio substitutes the assay floor for a missing off-target (above): $\mathrm{p}K_d = 5.0$ on Davis and Klaeger, $\mathrm{p}K_i = 4.0$ on Metz, and 0% inhibition on Anastassiadis. The S-score is defined for every compound. This yielded 30 rank vectors per dataset for Davis, Klaeger, and Metz (11 for the S-score, 7 for entropy, 7 for Gini, and 5 for ratio), and 28 for Anastassiadis, whose coarser native percent-inhibition scale supports only 9 S-score threshold values rather than 11.

For each compound, rank instability was quantified as the standard deviation of its rank across all configurations for that dataset (overall instability) and separately across parameter configurations within each definition family (within-family instability). Cross-definition-family agreement was assessed using Spearman rank correlation between the median rank vectors of each definition family. The median across parameter values was used because it is less sensitive to outlying configurations than the mean. Using mean ranks instead changes no cell of Table 1 by more than 0.04 on Davis and Klaeger, and by at most 0.075 on the other two datasets.

Associations between rank instability and binding profile features were assessed using Spearman correlation on Klaeger and Davis. Both report $\mathrm{p}K_d$, so the features are on the same scale and each relation found on one can be tested on the other. Profile features computed for each compound included: the number of active kinases ($\mathrm{p}K_d > 6$), the gap between the top and second-ranked affinities, the standard deviation and range of active affinities, the

absolute affinity of the second-ranked target, and the skewness of the active affinities. The first two were also tested on Anastassiadis and Metz, against entropy and ratio instability respectively (Results). All correlations are reported with two-tailed p-values, uncorrected for multiple testing. Significance thresholds of $p < 0.05$, $p < 0.01$, and $p < 0.001$ are denoted *, **, and *** respectively. Analyses were performed in Python 3 using NumPy, pandas, and SciPy.[22–24] A secondary analysis relating selectivity rankings to clinical adverse-event outcomes is summarized in the Limitations section. The analysis code and full per-definition results are available in the code repository (see Data and Software Availability), and Supporting Information Table S6 lists the script that computes each reported result.

Motivated Properties

The instability patterns reported in Results motivate four properties. Each is stated formally below, together with the failure mode it targets and how a measure is assessed against it. We evaluate the four definitions and the candidate measure against them (Results). Let $\mathbf{x} = (x_1, \ldots, x_N) \in \mathbb{R}^N$ denote a compound's binding profile across N kinases, where $x_i = \mathrm{p}K_{d,i}$ and N is the panel size, which varies across datasets and subpanels. A selectivity measure is a function $f : \mathbb{R}^N \to \mathbb{R}$, defined for every panel size N, where higher values indicate greater selectivity. A ranking over a set of compounds is defined by ordering their f-scores. We write x_0 for the assay's detection floor and τ^* for the activity threshold ($\mathrm{p}K_d$ or $\mathrm{p}K_i = 6$; 50% inhibition on Anastassiadis).

D1 (reliability threshold). $f(\mathbf{x})$ is undefined, or explicitly flagged as unreliable, whenever $\max_i x_i \leq \tau^*$.

A compound with no binding above the activity threshold has no target on which to ground a selectivity judgment. Its rank then depends on how each measure handles such profiles (Type 1, Results). D1 is a reporting rule and is assessed as such.

D2 (bounded gap sensitivity). For any $\mathbf{x}$ with $\max_i x_i > \tau^*$, if f depends on a choice of reference off-target indexed by k, there must exist g with $g(0) = 0$ such that

$|f(\mathbf{x}; k) - f(\mathbf{x}; k+1)| \leq g(x_{(k)} - x_{(k+1)})$.

When the k-th and $(k+1)$-th ranked affinities are nearly equal, moving the reference off-target past the $(k+1)$-th kinase (from k to $k+1$) must change the score by only a small amount. This targets the ratio's near-tie instability (Type 2, Results) and is assessed from each measure's functional form. A measure with no reference-off-target parameter satisfies D2 vacuously.

D3 (distributional consistency). Let $B = [x_0, \tau^*]$, the interval between the assay's detection floor and the activity threshold, where an analyst must decide how much sub-threshold binding counts. Let θ be a parameter of f that an analyst must choose on the affinity scale (the S-score's τ, the entropy's and Gini's β). A pair of compounds $\mathbf{x}, \mathbf{y}$ *reverses in* B if $f(\mathbf{x}; \theta) - f(\mathbf{y}; \theta)$ is positive at some $\theta \in B$ and negative at another. D3 requires that at most a fraction ε of the pairs of compounds that D1 allows to be scored reverse in B.

We use $\varepsilon = 1\%$. D3 targets the baseline sensitivity of compounds with few active kinases (Results) and is assessed empirically by computing the rankings across B on each dataset. A measure with no such parameter satisfies D3 vacuously.

D4 (monotonicity under weak off-target addition). For any $\mathbf{x}$ with $\max_i x_i > \tau^*$, if $\mathbf{x}' = (x_1, \ldots, x_N, x_{N+1})$ with $x_{N+1} \leq \tau^*$, then $f(\mathbf{x}') \leq f(\mathbf{x})$.

That is, adding a sub-threshold binding interaction must not increase a compound's score. Otherwise, a compound can be made to look more selective by profiling it against more kinases it barely binds. D4 is assessed from each measure's functional form.

A Candidate Measure

The candidate measure is introduced to show that the four properties can be met together. It need not outperform existing measures on any particular dataset. For a profile $\mathbf{x}$ with $\max_i x_i > \tau^*$ (undefined otherwise, satisfying D1), define

$$w_i = \mathbf{1}[x_i > x_0]\, T \log\!\big(1 + e^{(x_i - \beta)/T}\big), \qquad p_i = \frac{w_i}{\sum_j w_j}, \qquad f(\mathbf{x}) = \sum_i p_i \log_2 p_i, \tag{5}$$

where x_0 is a constant representing the assay detection floor ($\mathrm{p}K_d = 5.0$ on Davis and Klaeger, $\mathrm{p}K_i = 4.0$ on Metz, and a 20% inhibition noise level on Anastassiadis), β is a smooth emphasis baseline (set to x_0 in the results below), and T is a smoothing width in the native units of the profile. Our conversion places non-detection on Anastassiadis at 0% inhibition, the floor used in the rank-stability analysis, but 54% of its entries lie between 0% and 20%, so a gate at 0% would admit assay noise. We set $T = \tau^* - x_0$, the distance from the floor to the reliability threshold: 1 $\mathrm{p}K_d$ unit on Davis and Klaeger, 2 $\mathrm{p}K_i$ units on Metz, and 30 percentage points on Anastassiadis, whose native scale is 0–100% inhibition rather than log-affinity. The function f is the negative Shannon entropy of the smoothed, normalized profile, i.e. the negative logarithm of the effective number of targets the compound binds.

The candidate is a gated, smooth-hinge variant of the selectivity entropy and differs from it in three ways, one for each property the entropy fails: (1) an explicit reliability gate, for D1; (2) a non-binder gate fixed at the floor x_0, so that no free parameter sets the activity boundary, for D3; and (3) a smooth softplus hinge above x_0 in place of the entropy's hard cutoff $\max(x_i - \beta, 0)$, for D4. The hard cutoff gives a kinase just above β an arbitrarily small weight, which is what lets a sub-threshold addition raise the entropy's score. The softplus gives every gated kinase a weight of at least $T \ln 2$, which rules this out:

Proposition. With $\beta = x_0$ and $T \geq (\tau^* - x_0)/\ln(2^e - 1) \approx 0.58\,(\tau^* - x_0)$, adding a kinase with affinity at most τ^* never raises the candidate's score.

Proof. A kinase at or below x_0 has zero weight and leaves the score unchanged. Otherwise, every gated kinase, the added one included, has weight above $m = T \ln 2$. Since the softplus increases with affinity, an added kinase with affinity at most τ^* has weight $a \leq T \ln(1 + e^{(\tau^* - x_0)/T})$, and, by the assumption on T in the proposition, this is at most $e\,m$.

Let the original profile have $n \geq 1$ gated kinases ($n \geq 1$ because D1 scores it only if some kinase lies above $\tau^* > x_0$), total weight $W > nm$, and entropy H (in nats), and let $q = a/(W + a)$ be the share of the total weight carried by the added kinase. Once that kinase is added, the entropy is $(1 - q)H + h(q)$, where $h(q) = -q \ln q - (1 - q) \ln(1 - q)$ is

the binary entropy. So, the entropy does not decrease if and only if $H \leq h(q)/q$. Because h is concave with $h(0) = 0$, $h(q)/q$ decreases in q. Since $q < r/(n+r)$ with $r = a/m \leq e$, this gives $h(q)/q > \ln \frac{n+r}{r} + \frac{n}{r} \ln \frac{n+r}{n}$. As $H \leq \ln n$, it suffices that $\ln n$ is at most this bound, which rearranges to $\ln r \leq (1 + \frac{n}{r}) \ln(1 + \frac{r}{n})$. The right-hand side is at least 1 for all $n, r > 0$, because $\ln(1+t) \geq t/(1+t)$, and $\ln r \leq 1$. The entropy therefore does not decrease, so the score (its negative) does not increase. □

The choice $T = \tau^* - x_0$ satisfies this condition on every dataset. With this T, the same argument covers every $\beta \in B$: the lightest gated kinase then has weight above $m = T \ln(1 + e^{(x_0 - \beta)/T})$, and $a/m \leq 2.22 < e$.

Results

Selectivity Definitions Cluster into Two Families

Across the 30 parameter configurations (Methods), compound rankings were computed for all drugs in both datasets. Spearman correlations between the median ranking of each definition family reveal a consistent two-cluster structure (Table 1). On the Davis dataset, S-score, entropy, and Gini are mutually correlated ($r = 0.946$–0.999), while ratio correlates substantially less with each ($r = 0.34$–0.46). On the larger Klaeger dataset, the same structure is preserved but with lower within-cluster correlations ($r = 0.72$–0.91). The lower values come from the 16 compounds with no active kinase, whose scores rest on sub-threshold values alone (Type 1, below). Among the 206 compounds with an active kinase, the within-cluster correlations are 0.91–0.998, close to Davis's. The ratio definition is the consistent outlier in both datasets, with cross-family correlations of $r = 0.22$–0.62. On Klaeger, we tested whether the Gini sits closer to the S-score than to the ratio by bootstrapping the difference between its two correlations (1,000 resamples of compounds with replacement). With all compounds included, the difference (0.72 against 0.62) is not significant (95% interval $[-0.06, 0.26]$). Among the 206 compounds with at least one active kinase, it is significant (0.91 against 0.63,

interval $[0.19, 0.37]$).

Table 1: Spearman correlations between median rankings by definition family. Upper triangle: Davis ($n = 68$). Lower triangle: Klaeger ($n = 222$).

	S-score	Entropy	Gini	Ratio
S-score	—	0.946	0.952	0.456
Entropy	0.910	—	0.999	0.343
Gini	0.722	0.840	—	0.343
Ratio	0.223	0.455	0.616	—

This two-family structure follows from what the definitions measure. The S-score, entropy, and Gini all quantify the concentration of activity across the full kinase panel, and the entropy and Gini are computed from the same baseline-subtracted profile. The ratio quantifies the gap between the primary target and one off-target and ignores the rest of the profile.

The Two-Family Structure Replicates Across Assay Technologies

The Davis and Klaeger pattern of Table 1 extends to the two further datasets, across three assay technologies and both readout types (Figure 2). In all four, the S-score, entropy, and Gini remain mutually correlated (within-family Spearman $r = 0.72$–0.999) while the ratio definition is the consistent outlier. Its correlation with the distribution-based family is 0.49–0.56 on the Anastassiadis functional-activity dataset and only 0.14–0.19 on the Metz pK_i dataset. On Anastassiadis, the separation comes mainly from the 44 compounds with no active kinase (Type 1, below). Among the other 134, the ratio's correlation with the family rises to 0.79–0.83, against 0.87–0.999 within it. On Metz the low agreement partly reflects coverage (Methods).

The instability findings also replicate. Pooled across the definitions' sweeps, compounds with no active kinase have higher rank instability than active compounds wherever they occur (mean rank $\sigma = 36.8$ vs. 21.7 on Anastassiadis and 178.5 vs. 108.1 on Metz; for Klaeger see D1, below; Davis has none). On each dataset the contrast comes mainly from the definitions disagreeing with one another rather than from each definition's own parameter sweep (Type 1,

below). Within a single sweep, it is significant for the Gini on Klaeger and Anastassiadis, and for the entropy and the ratio on Anastassiadis. On Metz, it reverses for the S-score, entropy, and Gini: the five zero-active compounds have lower rank instability than active compounds ($\sigma = 7.8$ vs. 71.2, 11.0 vs. 24.5, and 5.4 vs. 24.2, respectively, all $p < 0.05$). The number of active kinases negatively predicts entropy instability in every dataset ($r = -0.28$ to -0.42; on Davis only nominally, $p = 0.02$). The top$_1$–top$_2$ affinity gap predicts ratio instability on Davis, Klaeger and Metz ($r = -0.44$, -0.34, and -0.27, all $p < 0.001$). It also predicts instability on Anastassiadis once the compounds saturated at the 100% inhibition ceiling are excluded (below).

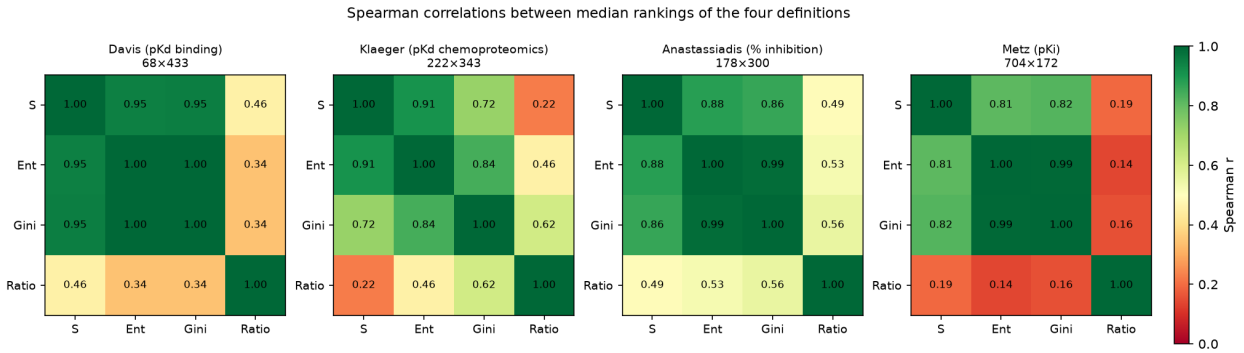


Figure 2: Spearman correlations between the median rankings of the four selectivity definitions on each dataset (compounds × kinases above each panel). S: S-score; Ent: entropy.

Rank Instability Is Structured and Associated with Profile Shape

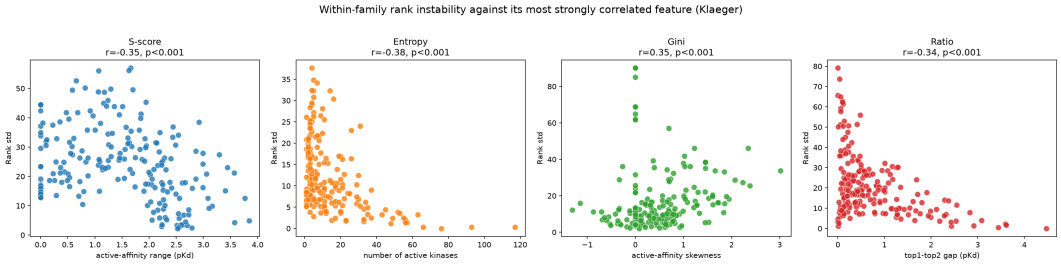


Figure 3: Within-family rank instability against the profile feature most strongly correlated with it, for each definition family (Klaeger, compounds with an active kinase, $n = 206$): active-affinity range (S-score), number of active kinases (entropy), active-affinity skewness (Gini), and top$_1$–top$_2$ gap (ratio). Spearman r above each panel.

Overall rank instability varies substantially across compounds (Klaeger: mean $\sigma = 34.2$, range 7.0–92.6; Davis: mean $\sigma = 8.5$, range 1.8–19.3). Instability is not uniformly distributed: within each definition family, specific binding profile features are associated with which compounds receive unstable rankings (Table 2).

Within the ratio family, the gap between the top and second-ranked affinities (top_1–top_2 gap) is the strongest predictor of instability on both datasets (Figure 3; Type 2, below).

Within the entropy family, the number of active kinases ($\text{p}K_d > 6$) negatively predicts instability (Table 2; for the Gini, see below). Compounds with many active kinases have broad profiles whose entropy scores are stable across baseline choices. Compounds with few active kinases are sensitive to the baseline because small changes can move targets in or out of the active set, altering the distribution substantially. Consistent with this, the share of a compound's above-floor kinases that cross some baseline in the sweep correlates with its entropy instability ($r = 0.19$, $p = 0.007$ on Klaeger). The active affinity range is also a significant predictor for the S-score and entropy on Klaeger ($r = -0.35$ and -0.27, both $p < 0.001$). For the Gini it is not significant ($r = -0.10$, $p = 0.14$). Range rises with the number of active kinases (Spearman 0.75). Holding that number fixed, range still predicts S-score instability (partial $r = -0.23$, $p < 0.001$) and no longer predicts entropy instability ($r = 0.02$, $p = 0.74$).

Gini instability's strongest predictor on Klaeger ($r = +0.346$, $p < 0.001$) is skewness. Unlike the other features above, it targets what Gini measures directly, i.e. the shape of the active-affinity distribution's inequality. Right-skewed active distributions are the least stable. In such profiles most active kinases sit just above the threshold, with one or a few much more potent outliers, so a shift in β moves much of the weight in or out of the active set. Among the 161 Klaeger compounds with at least three active kinases, skewness correlates with the share of a compound's weight carried by kinases that the β sweep crosses ($r = 0.45$), and that share predicts Gini instability with the number of active kinases held fixed (partial $r = 0.71$). Holding it fixed as well lowers the skewness association from 0.57 to 0.31, so this

mechanism accounts for part of the association. As with active affinity std and range above, the relationship does not replicate on Davis ($r = +0.043$, $p = 0.73$).

Unlike the other families, the S-score's instability is significantly predicted by every profile-shape feature except the top_1–top_2 gap (Table 2), and most strongly by the spread of active affinities. This S-score pattern is specific to Klaeger: on Davis, no profile feature significantly predicts S-score instability (all $|r| \leq 0.16$, Table 2, right). The difference does not come from the two datasets containing different compounds. On the 46 compounds profiled in both, active affinity range predicts S-score instability on Klaeger ($r = -0.41$, $p = 0.01$, for the 37 with at least two active kinases) and not on Davis ($r = -0.10$, $p = 0.53$), and the same compounds have a median of 5 active kinases on Klaeger and 50 on Davis. Three relations in Table 2 are significant on both pK_d datasets: the ratio's dependence on the top_1–top_2 gap and on the second-ranked affinity, and the entropy's dependence on the number of active kinases. With a Bonferroni correction for the 24 tests per dataset, only the first remains significant on both. On Anastassiadis the ratio–gap dependence fails ($r = -0.104$, $p = 0.23$) because of eight compounds whose two best-inhibited kinases are both at 99.5% inhibition or more: the single-concentration readout saturates at 100%, so their top_1–top_2 gap is unresolved. Excluding them, the dependence holds there too ($r = -0.275$, $p = 0.002$). For the Gini, the dependence on the number of active kinases is significant on Davis ($r = -0.300$) and not on Klaeger ($r = -0.093$).

Table 2: Spearman correlations between binding profile features and within-family rank instability, on the Klaeger ($n = 206$ active drugs) and Davis ($n = 68$) datasets. Columns are the S-score (S), entropy (E), Gini (G), and ratio (R) families. Significance: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

	Klaeger ($n = 206$)				Davis ($n = 68$)			
Feature	S	E	G	R	S	E	G	R
n_{active}	-0.272^{***}	-0.381^{***}	-0.093	-0.040	-0.153	-0.279^{*}	-0.300^{*}	$+0.294^{*}$
top_1–top_2 gap	$+0.039$	$+0.082$	$+0.031$	-0.341^{***}	-0.079	-0.007	-0.029	-0.444^{***}
Active affinity std	-0.281^{***}	-0.174^{*}	-0.117	-0.107	$+0.009$	-0.015	$+0.005$	$+0.040$
Active affinity range	-0.354^{***}	-0.271^{***}	-0.102	-0.116	$+0.011$	-0.111	-0.098	$+0.134$
top_2 pK_d	-0.314^{***}	-0.200^{**}	-0.192^{**}	$+0.186^{**}$	-0.055	-0.203	-0.187	$+0.296^{*}$
Active affinity skew	-0.162^{*}	-0.028	$+0.346^{***}$	-0.108	$+0.026$	$+0.066$	$+0.043$	$+0.069$

The skewness of a compound's active affinities is undefined when it has fewer than three active kinases, and so are the standard deviation and range of those affinities when the compound has fewer than two. Table 2 assigns these compounds the value 0 (the vertical clusters at zero in Figure 3). Recomputing each correlation over only the compounds for which the feature is defined (161 and 180 of the 206 active Klaeger compounds, respectively) strengthens the Klaeger associations reported above, including Gini–skewness ($r = +0.482$) and the S-score's and entropy's dependence on active affinity range ($r = -0.479$ and -0.396, all $p < 0.001$). It also makes the Gini's and the ratio's weak dependence on active-affinity spread nominally significant ($|r| = 0.18$–0.28, $p < 0.05$). The one exception is the S-score's weak dependence on skewness, which falls from $r = -0.16$ ($p = 0.02$) to -0.15 ($p = 0.05$). On Davis, where only three compounds have fewer than three active kinases, no correlation changes materially.

Five additional candidate features were tested: two concentration measures from industrial-organization economics ((1) the top-target concentration ratio (CR_1) and (2) the Herfindahl–Hirschman Index (HHI)). These were sourced from the same field as the Gini coefficient.[6] We also tested (3) the raw primary-target affinity (top_1 pK_d), (4) a near-tie count generalizing the top_1–top_2 gap, and (5) the active-affinity coefficient of variation. CR_1 and HHI turned out to be near-collinear with n_{active}, which is already in the table ($r = -0.98$ to -0.99 and -0.998, respectively, on both datasets), and thus they carry essentially the same information. The remaining three were weaker than the strongest feature already in the table, in every family and on both datasets.

Three Mechanistically Distinct Sources of Instability

Examination of the most and least unstable compounds in the Klaeger dataset reveals three structurally distinct sources of rank instability, each implicating a different aspect of how the definitions are constructed. Together they account for all 20 most unstable Klaeger compounds. This breaks down as: 16 have no active kinase (Type 1), 2 have a top_1–top_2 gap

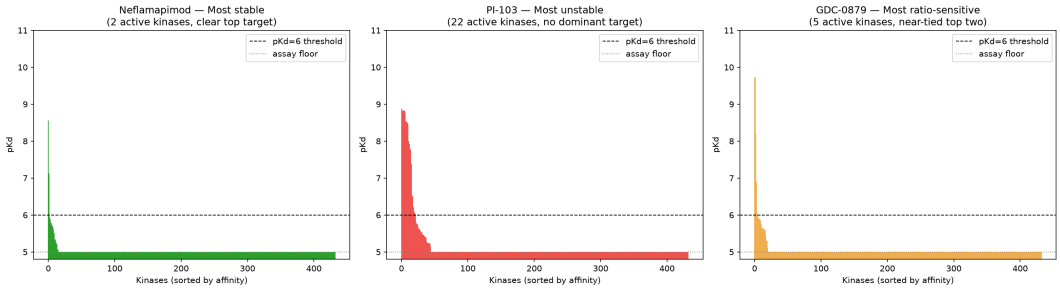


Figure 4: Binding profiles of three Davis compounds, kinases sorted by affinity. Left: neflamapimod (VX-745), most stable ($\sigma = 1.8$), two active kinases. Center: PI-103, most unstable ($\sigma = 19.3$), 22 active kinases with no dominant target. Right: GDC-0879, most ratio-sensitive (Type 2), five active kinases including two near-tied top targets ($\Delta \mathrm{p}K_d = 0.023$). Dashed line: $\mathrm{p}K_d = 6$ activity threshold; dotted line: $\mathrm{p}K_d = 5$ assay floor.

below 0.1 (Type 2), and 2 have at least 10 active kinases and a gap of at least 1 (Type 3). Figure 4 shows example profiles from Davis (the extremes are easier to see on the complete matrix there).

Type 1: Zero-active compounds. The five most unstable Klaeger compounds (AZD-8055, BMS-911543, Vatalanib, Filgotinib, ARRY-380; rank $\sigma > 78$) all have zero active kinases at the $\mathrm{p}K_d > 6$ threshold in the Klaeger panel. This reflects the assay result: all five are established, potent inhibitors of specific kinase targets (mTOR, JAK2, VEGFR2, JAK1, and HER2, respectively), none of which appear as high-confidence sub-micromolar hits in the Klaeger chemoproteomic screen. For three of them (AZD-8055, vatalanib, ARRY-380), the target (mTOR, KDR, ERBB2) was not quantified for any of the 229 compounds with a fitted target in the Kinobeads table. This is consistent with low expression or poor capture of those kinases in the cell lysates. For the other two, this explanation does not hold: JAK2 and JAK1 were quantified for 7 and 12 other compounds (up to $\mathrm{p}K_d = 8.3$), yet no binding was detected for BMS-911543 or filgotinib.

All four definitions nonetheless score these compounds from sub-threshold values alone, and they disagree about them. The S-score and entropy rank them among the most selective (median ranks 22 and 25 of 222), and the ratio in the bottom third (156). The Gini ranks them near the top where its score is defined, and near the bottom where the convention for

undefined scores applies (Methods; mean ranks 35 and 214). Their pooled rank instability is therefore high (D1, below). Davis has no such compounds.

Type 2: Near-tied top targets. When an active compound's two best-bound kinases have near-identical affinities, the ratio depends heavily on k. $R(1)$ compares the two near-tied kinases and is close to zero, and $R(2)$, which skips past the tie, can be large (D2, below). GDC-0879 in the Davis dataset (top_1–top_2 gap = 0.023 pKd units) is the extreme case, with the most unstable ratio ranking on Davis.

Type 3: Broad profiles, often with a dominant primary target. Among active compounds, the ten largest entropy-ratio disagreements in the Klaeger dataset all involve broad profiles with 14–33 active kinases. Half of them also have a large top_1–top_2 gap, for example Dovitinib ($n_{\text{active}} = 26$, gap = 1.935) and BMS-690514 ($n_{\text{active}} = 20$, gap = 1.985). These compounds have one clearly dominant target and also bind many kinases at moderate affinity. The ratio ranks them among the most selective compounds (ranks 22–60 of 222 for the five such compounds in the top ten), while the entropy and Gini rank them among the least selective (ranks 157–191).

Which compounds are most stable differs between datasets. On Klaeger the most stable compounds are among the most promiscuous. XL-228 ($\sigma = 7.0$) and PF-03814735 ($\sigma = 7.8$), with 55–57 active kinases, have median ranks of 208–220 of 222 under every definition, and mean instability falls from about 31–33 for compounds with 1 to 40 active kinases to 16 above 40. On Davis compounds with few active kinases are the most stable on average (mean $\sigma = 3.1$ for one or two, 4.4 for three to five). The most stable, neflamapimod (Figure 4, left, $\sigma = 1.8$), has two, with median ranks of 3–5.5 under the four definitions. The 57 Davis compounds with more than 20 active kinases are less stable (mean $\sigma = 9.1$, against 5.3 for the other 11). The S-score, entropy, and Gini give the five broadest median ranks of 63–68 of 68, and the ratio gives them 29–52. Without the ratio's configurations, the mean instability of the 57 broad compounds falls from 9.1 to 5.1.

Panel Size Dependence of Selectivity Rankings

Karaman et al.[4] scored each compound's S-score on 500 random subpanels of each size and found that it varies more as panels shrink, concluding that a single small panel is unlikely to rank compounds reliably by overall selectivity. We test that conclusion by measuring how well each definition's ranking on a random subpanel agrees with its full-panel ranking, done for all four definitions on all four datasets.

For each panel size, in steps of 30 from $p = 50$ kinases up to the full panel (433 kinases on Davis, 343 on Klaeger, 300 on Anastassiadis, and 172 on Metz), we drew 50 random subsets of p kinases from the full panel, computed each selectivity definition on the subpanel, and measured agreement with the full-panel ranking using Spearman correlation. Each definition uses the dataset's detection floor x_0 as its baseline (entropy, Gini) or fallback (ratio, $k = 1$), and the activity threshold as the S-score cutoff. On Anastassiadis, x_0 is set to the 20% noise level used for the candidate (Methods). We call a panel size "recovering" the full-panel ranking once this mean correlation exceeds 0.90, a bar borrowed from the "excellent" range of intraclass-correlation guidelines.[25] Changing the bar within $[0.80, 0.95]$ changes the p^* values but never reverses the order of two definitions on any dataset (Supporting Information, Table S1).

The four definitions differ substantially in their panel size requirements (Figure 5). Selectivity entropy and the S-score recover the full-panel ranking fastest on every dataset: at 50 kinases, the smallest panel tested, on Davis and Anastassiadis, at 80 (entropy) and 110 (S-score) on Klaeger, and at 80 (entropy) and 50 (S-score) on Metz. The ratio is the slowest everywhere, needing 410 of 433 kinases on Davis, 290 of 300 on Anastassiadis, 170 of 172 on Metz, and the complete panel on Klaeger, where it reaches only $r = 0.888$ at 320 kinases. The Gini coefficient falls in between: it needs 110 kinases on Davis, 200 on Anastassiadis, and 290 on Klaeger, but matches the entropy on Metz (80). Its slower recovery disappears when the comparison is restricted to compounds with an active kinase on both the subpanel and the full panel (The Candidate Measure, below). The Metz values carry the coverage

caveat of Methods.

Kinase profiling panels used in practice vary in size, from focused subpanels to near-kinome panels. On a 50-kinase subpanel, entropy- and S-score-based rankings are reasonably stable on every dataset ($r = 0.81$–0.97). Ratio rankings are nearly uncorrelated with the full-panel ranking ($r = -0.08$ to 0.18), and Gini rankings range from close agreement on Metz ($r = 0.87$) to none on Klaeger ($r = -0.09$). The practical consequence is that the smaller the panel, the more the choice of definition determines the answer.

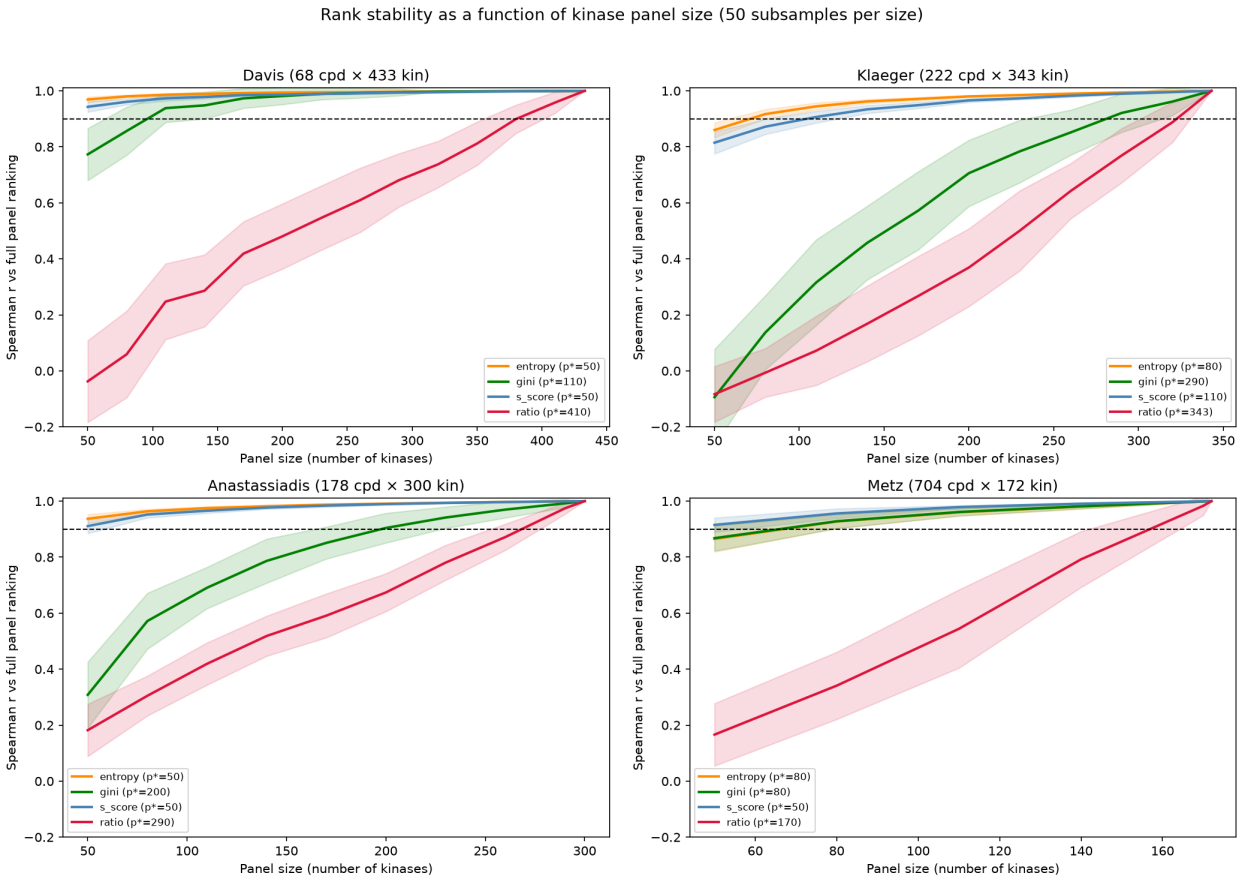


Figure 5: Mean Spearman correlation between rankings computed on random kinase subpanels and rankings on the full panel, as a function of subpanel size, on all four datasets. Shaded bands indicate ± 1 standard deviation across 50 subsamples; the dashed line marks $r = 0.90$, and p^* (legend) is the smallest panel reaching mean $r > 0.90$.

Existing Definitions Against the Properties

Table 3 summarizes how the four definitions and the candidate measure fare against the properties D1–D4 defined in Methods. None of the four definitions satisfies all four, and D1 is the only property all four violate.

Table 3: Satisfaction of the four motivated properties by existing selectivity definitions and by the candidate measure. ✓: satisfied substantively. (✓): satisfied vacuously. ✓†: satisfied only with the activity boundary fixed at the assay floor x_0. ×: violated. Non-vacuous D3 verdicts are empirical (four datasets); the others follow from each measure's reporting rule or functional form.

Property	S-score	Entropy	Gini	Ratio	Candidate
D1 (reliability threshold)	×	×	×	×	✓
D2 (bounded gap sensitivity)	(✓)	(✓)	(✓)	×	(✓)
D3 (distributional consistency)	×	×	×	(✓)	✓†
D4 (monotonicity, weak off-target)	×	×	×	✓	✓

D1. None of the four definitions has a reliability condition, so each assigns a score to the 16 Klaeger compounds with no binding above the activity threshold. Pooled across the four definitions' parameter sweeps, their rank instability is $\sigma = 74.0$ against 31.2 for active compounds ($p < 0.001$, Mann–Whitney U test; Figure 6, left). Within a single definition the contrast is significant only for the Gini ($\sigma = 86.1$ versus 15.9). It is small and not significant for the entropy (14.1 versus 11.9, $p = 0.25$) and the ratio (24.0 versus 20.7, $p = 0.08$), and reverses for the S-score (18.5 versus 24.3, $p = 0.07$), whose score is exactly 0 for every such compound at the nine thresholds at or above τ^*. The pooled contrast comes instead mainly from definitions disagreeing about these compounds (Type 1, above): it persists with the undefined entropy and Gini configurations dropped ($\sigma = 59.6$ against 30.6) and with only the S-score and ratio pooled (66.0 against 36.7, both $p < 0.001$). The Gini's contrast results from the convention for undefined scores (Methods). A zero-active compound's entropy and Gini are defined at only 1–4 of the 7 baselines swept (mean 3.2). When instability is computed only over each compound's defined configurations, ranking compounds at each baseline among those defined there, zero-active compounds are *more* stable than active ones

(entropy: $\sigma = 9.8$ vs. 12.8, $p = 0.046$; Gini: $\sigma = 9.5$ vs. 13.0, $p = 0.028$). This uses 14 of the 16 compounds, since the other two are defined at fewer than two baselines. The case for D1 is therefore not that these scores are unstable within a definition, but that they rest on sub-threshold values alone, which each definition treats differently. Any of the four definitions could satisfy D1 by withholding or flagging such scores. For the ratio, the violation comes from our implementation's floor fallback (Methods, "Ratio").

D2. The ratio violates D2 when its top two affinities are nearly tied. On the 17 Klaeger compounds with a top_1–top_2 gap below 0.05, mean $R(1) = 0.02$ (the near-tied second kinase is treated as the off-target) against mean $R(2) = 0.42$ (it is treated as a co-target). This is the Type 2 instability above. The S-score, entropy, Gini, and candidate have no reference off-target and satisfy D2 vacuously. The S-score's sensitivity to kinases near its threshold is a D3 matter.

D3. Swept over $B = [x_0, \tau^*]$, the S-score, entropy, and Gini each reverse the order of 8–23% of pairs of scored compounds, on every dataset (S-score 7.9–20.6%, entropy 8.8–22.6%, Gini 9.1–22.4%; highest on Metz), far above $\varepsilon = 1\%$. The candidate's emphasis baseline reverses at most 0.72%, so every verdict in Table 3 is the same for any ε from 1% to 7%. Making β an explicit parameter is what makes this testable: the published originals leave it implicit ($K_a \to 0$ for the entropy, 0% inhibition for the Gini). The ratio has no baseline and satisfies D3 vacuously.

D4. The S-score and Gini violate D4 by construction: both are computed over the full panel of size N, so adding a kinase the compound does not bind lowers the S-score's hit fraction and raises the Gini's inequality, inflating the apparent selectivity. This matters whenever panels differ in size or composition, as between Davis and Klaeger. The ratio satisfies D4, because a sub-threshold kinase cannot displace the primary target and cannot lower the reference off-target's affinity. The entropy satisfies D4 only for additions at or below its baseline β ("Why the Smooth Hinge Matters", below).

The Candidate Measure

The candidate measure (Methods) is correctly oriented: its score falls with the number of active kinases on every dataset (Spearman -0.783 to -0.969). The non-binder gate is what makes this so. Without it, the non-binding kinases at the floor dominate the profile and the score inverts, rising with the number of active kinases ($r = +0.93$ on Klaeger).

Because the candidate is a local modification of the entropy, the two rank the compounds they both score almost identically (full-panel Spearman 0.994–0.999 on all four datasets). The properties do not act by reordering compounds in the body of the distribution. They act on which compounds receive a score at all (D1, which flags 16 of 222 Klaeger compounds), on whether the ranking survives a shift in an analyst's parameter (D3), and on whether adding a sub-threshold kinase can raise a score (D4).

The candidate satisfies D1 by its reliability gate and D2 vacuously. For D3, its only free parameter is the emphasis baseline β, and sweeping it over B reverses 0.43–0.72% of pairs, so it passes on every dataset. Its activity boundary is fixed. Sweeping the gate x_0 itself over B reverses 8.5–22.5% of pairs, against 8.8–22.6% for the entropy's baseline, and the two sweeps produce similar per-compound instability on Klaeger (mean rank $\sigma = 7.2$ and 7.5; Figure 6, right). The candidate's D3 standing therefore comes entirely from declaring x_0 a fixed constant, and Table 3 marks it $\checkmark^{\dagger}$. D4 holds by the proposition in Methods. It also holds on every compound in all four datasets when one kinase is added at τ^*, which is the strongest binding that D4 allows (i.e. the addition most likely to raise the score; Supporting Information Table S4).

A usable measure must also be recoverable from realistic panels. With the panel-size subsampling analysis above, the candidate reaches mean Spearman $\rho > 0.90$ with its full-panel ranking at $p^* = 50$ kinases on Davis and Anastassiadis (the smallest panel tested) and 80 on Klaeger and Metz (Figure 7). These values match the entropy on every dataset. These p^* values apply the formula without the D1 gate, so they include compounds the gate would refuse to score (at 50 kinases, 9% of Davis compounds, 42% of Klaeger, 43% of Anastassiadis,

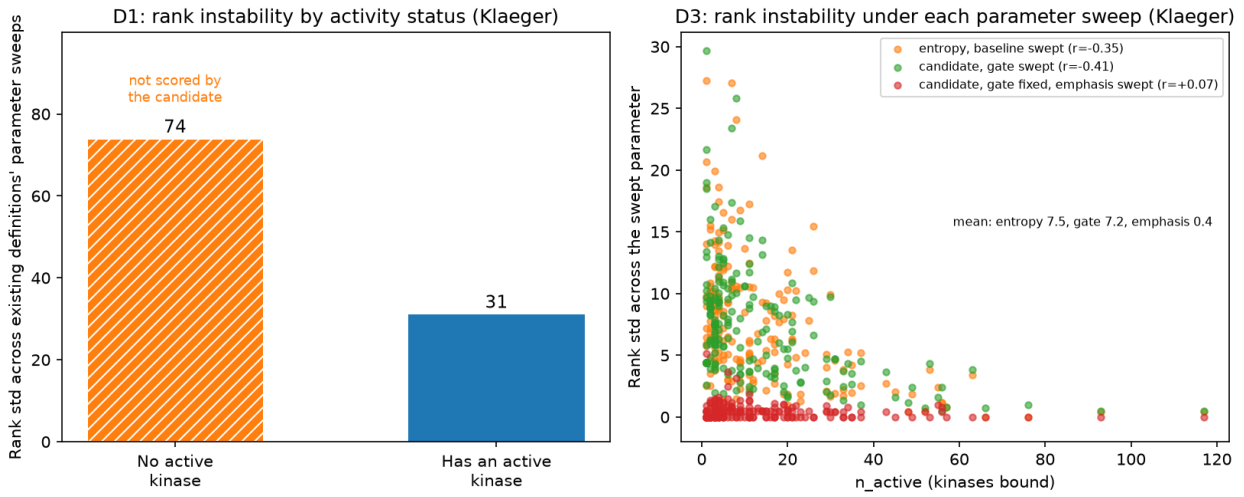


Figure 6: Rank instability on Klaeger. *Left (D1):* rank standard deviation across the existing definitions' parameter sweeps, for compounds with no binding above the activity threshold and for compounds with at least one active kinase. *Right (D3):* per-compound rank standard deviation against the number of active kinases, with one parameter swept over $B = [5.0, 6.0]$: the entropy's activity baseline (orange), the candidate's gate x_0 with its emphasis baseline pinned to x_0 (green), and the candidate's emphasis baseline with x_0 fixed at the assay floor (red).

and 2% of Metz, counting those with no active kinase on the full panel). Restricting the comparison to compounds that D1 allows to be scored on both the subpanel and the full panel leaves the candidate's and the entropy's p^* unchanged and brings the Gini to the same values (50 on Davis and Anastassiadis, 80 on Klaeger and Metz). The Gini's gain comes from removing compounds whose Gini is undefined on the subpanel, which the convention ranks least selective (Methods). Excluding only these compounds gives the Gini the same p^* as the gate on every dataset.

The featured candidate is one member of a family of measures that satisfy D1–D4 on these data: a per-compound quantile floor in place of the fixed floor and the Rényi orders 0.5 and 1 do so on Davis and Klaeger, and smoothing widths from 0.75 to 2 times $\tau^* - x_0$ on Klaeger (D4 is proved only for the Shannon order, $\alpha = 1$). These variants, including their panel recoverability, are reported in Supporting Information (Section S5).

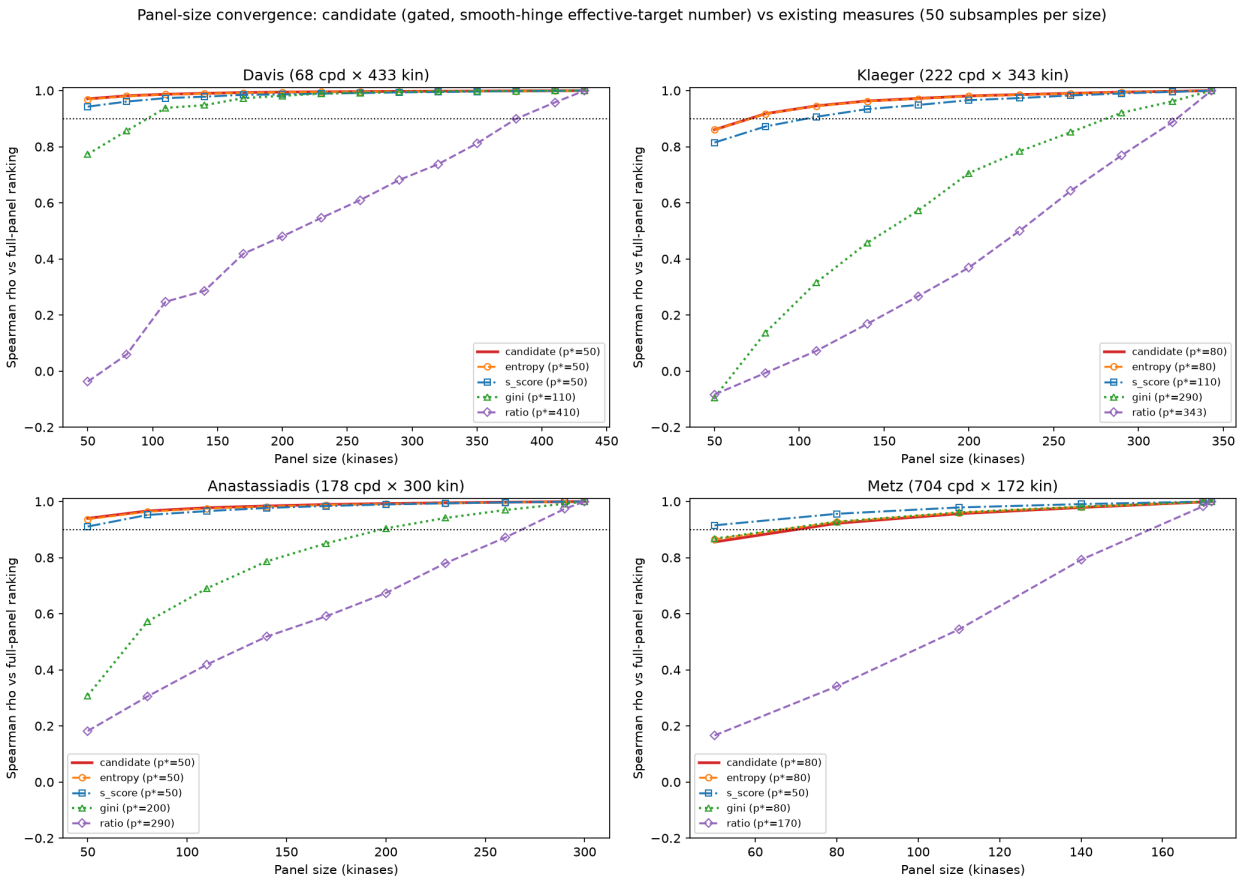


Figure 7: Panel-size convergence of the candidate measure compared with the four existing definitions, on all four datasets (50 random subpanels per size). Curves show the mean Spearman correlation between the ranking computed on a random subpanel and the full-panel ranking; p^* (legend) is the smallest panel reaching mean $\rho > 0.90$.

Why the Smooth Hinge Matters

The purpose of this subsection is to show that the candidate and the entropy differ in principle, although they rank these four datasets almost identically, and that the difference comes from the candidate's functional form. The property at stake is D4.

The entropy can violate D4. Its hard cutoff gives a kinase just above the baseline β an arbitrarily small weight. A profile with many such kinases therefore has high entropy, and adding one kinase that is bound more strongly, but still below τ^*, concentrates the distribution and raises the selectivity score. For example, with $\beta = 5.0$ and $\tau^* = 6.0$, a profile with one kinase at $\mathrm{p}K_d = 7.0$ and fifty at 5.1 has an entropy-based selectivity of -4.89 bits,

and adding one kinase at $pK_d = 6.0$ raises it to -4.83. On Davis, Klaeger, and Metz, adding one kinase at τ^* never raises the entropy's score for a compound with an active kinase, at any baseline in their sweeps. Compounds with no active kinase are left out because D1 withholds their scores.

On Anastassiadis it does at baselines below the 20% floor: weighting each kinase by its raw percent inhibition ($\beta = 0\%$), it raises the score for 34% of the 134 compounds with an active kinase, and for 16% at $\beta = 5\%$, although for none at 15% or above. Adding real kinases rarely does this: expanding random 50- or 100-kinase panels by 1 to 30 kinases that a compound binds below threshold raises the entropy's score in at most 0.1% of cases.

The candidate cannot violate D4. By the proposition in Methods, no sub-threshold addition can raise its score, and its condition on T is sharp: just below it, a profile with one strong target and enough kinases just above the floor violates D4. With $T = 1$ on Metz or $T = 10$ on Anastassiadis, both below the bound, constructed profiles of this kind violate D4 (by up to 0.014 bits on Anastassiadis), although none of the real compounds does. Setting $T = \tau^* - x_0$ (Methods) leaves Davis and Klaeger unchanged and changes the Metz and Anastassiadis rankings negligibly (Spearman 0.999 against $T = 1$ and $T = 10$, respectively).

Discussion

The central finding of this work is that existing kinase inhibitor selectivity definitions, placed on a common log-affinity scale, do not measure the same quantity. The ratio definition and the distribution-based family (S-score, entropy, Gini) produce substantially different compound rankings, and the largest disagreements are concentrated in identifiable kinds of profiles. This has practical consequences: a compound that ratio-based analysis designates as selective may be designated as moderately promiscuous by entropy-based analysis, and vice versa, depending on the shape of its binding profile. Among compounds that clear the activity threshold, the two families disagree most on broad profiles (Type 3, Results).

This coexistence may have a historical explanation. In classical pharmacology and analytical chemistry, selectivity is inherently pairwise. IUPAC formalizes it as a selectivity coefficient, the ratio of binding constants for an analyte and an interferent,[26] and ratio measures follow that tradition. Kinome-wide profiling, however, poses a distributional question: how concentrated is a compound's binding across hundreds of kinases? Entropy and Gini address this question directly.

Practical Recommendations

Pending a formal resolution, three practical recommendations follow from our findings. First, D1 (reliability threshold) should be adopted as a reporting standard. Compounds with no binding above the $pK_d > 6$ activity threshold should not receive selectivity scores, or should have their scores explicitly flagged as unreliable. Of the 222 Klaeger compounds, 16 (7.2%) fall into this category (Type 1). All 16 bind above the assay detection floor of $pK_d = 5.0$ (maximum affinities 5.16–6.00), so the gate sits at the activity threshold.

Second, selectivity assessments should report how rankings change with parameter choice, alongside the score at the chosen parameter value. Rank standard deviation across the parameter sweep is a compound-specific, interpretable quantity that identifies which compounds' assessments are robust to parameter choice, analogous to a confidence interval in statistical estimation. This rank standard deviation is not a guide to robustness across platforms: on the 46 compounds profiled in both Davis and Klaeger, it does not predict how far a compound's rank moves between the two ($|r| \leq 0.25$).

Third, when comparing selectivity across studies that use different definitions, the comparison should be restricted to the subset of compounds and parameter values for which the definitions agree. Within the distribution-based family, the S-score, entropy, and Gini correlate at $r \geq 0.72$ on the full panel (Table 1). They give similar rankings when parameters are chosen consistently and panels have similar, near-complete coverage. On smaller panels the Gini diverges on Davis, Klaeger, and Anastassiadis because of its convention for

undefined scores (Results). Ratio-based measures, in turn, should not be compared directly with distribution-based measures at any panel size without acknowledging the conceptual distinction.

Limitations

The analysis covers four profiling datasets spanning three assay technologies and two readout types, but these still represent a fraction of the chemical and kinase space relevant to drug discovery. The assays report on different scales and detect different shares of active interactions (Methods). The two-family structure (in our common log-affinity operationalization) replicates across all four datasets, and so, with the qualifications given in Results, do the two instability relations tested on all four (ratio–gap and entropy–n_{active}). This indicates that the findings are not artifacts of a single assay platform. On Anastassiadis, however, the two-family separation comes mainly from compounds with no active kinase, and on Metz it partly reflects coverage (Results). Findings should still be interpreted in light of these technological differences.

Our comparison is also restricted to four widely used, historically established definitions. Bosc et al.'s window score and ranking score[7] were not included. On the same four datasets both correlate strongly with the ratio (Spearman $r = 0.80$–0.96), so they do not form a third cluster. The CATDS score[10] was also not implemented directly, since it requires raw concentration-response data. A proxy computed from our baseline-subtracted affinity profiles correlates far more strongly with Gini and entropy than with $R(k)$ on all four datasets (Spearman 0.91–0.99 against Gini and 0.82–0.96 against entropy, against 0.45–0.70 for $R(k)$, Supporting Information Table S2). We did not add it as a fifth measure, because the distribution-based family is already represented by three members.

A further axis of platform dependence that this analysis does not examine is cell-based versus cell-free profiling. Recent work shows that selectivity classifications for the same compounds can differ substantially between biochemical binding panels and cellular target-engagement assays. For example, several type II (DFG-out-binding) inhibitors show cellular

interactions that cell-free assays miss.[27,28] A plausible contributor, not established by that work, is that the cellular assays report occupancy of full-length kinase in intact cells, with regulatory domains, binding partners, and intracellular ATP present. Many cell-free binding panels, instead, score isolated, recombinantly expressed kinase-domain constructs. This specific contrast does not map cleanly onto the two cell-free datasets used here. In particular, Davis profiles recombinant constructs (on the same competition-binding platform, Karaman et al. used catalytic domains for large multidomain kinases and full-length constructs for small single-domain ones[4]), while Klaeger's Kinobeads approach captures endogenous, full-length kinases directly from cell lysate.

We compared how much the choice of assay technology and the choice of selectivity definition each move a compound's selectivity ranking. We resolved 46 compounds profiled in both Davis (competition binding) and Klaeger (chemoproteomics) by matching PubChem CIDs against Klaeger's drug names, and computed each definition's median rank on this fixed set of 46 compounds (separately on each platform). Rankings agree less across platforms than across definitions, so the *platform effect* (same definition, Davis rank vs. Klaeger rank) is on average the larger of the two: Spearman $r = -0.06$ to 0.69 across the four definitions for the platform comparison (mean 0.44), against $r = 0.14$ to 1.00 for the within-platform, cross-definition comparison (mean 0.60). The ratio definition is the extreme case: its Davis ranking and its Klaeger ranking for the same 46 compounds are statistically indistinguishable from uncorrelated ($r = -0.06$, $p = 0.67$). This should temper any cross-study comparison of selectivity rankings regardless of which definition is used.

All four existing definitions and the candidate measure introduced above treat a compound's affinity vector as an unordered multiset. Thus, two compounds with identical affinity distributions receive identical scores regardless of which specific kinases carry the off-target activity. A compound whose off-targets are pharmacologically inert and one whose off-targets are known safety-relevant kinases (e.g. a kinase implicated in a class toxicity) are therefore scored identically if their affinity distributions match. None of the measures discussed here

incorporates target identity, and none accounts for exposure. A measured affinity alone does not establish whether a target is occupied in a patient, which also depends on the drug concentration achievable at a tolerated dose. A selectivity score is consequently, at best, a summary of *binding breadth*, and go/no-go decisions should weigh it alongside target-specific liability knowledge and exposure modeling.

We also examined whether in vitro selectivity predicts clinical toxicity, correlating the selectivity rankings of the FDA-approved Klaeger compounds with FAERS adverse-event rates ($n = 46$) and FDA-label discontinuation rates ($n = 33$). None of the rate-based outcomes were significantly associated with selectivity under any definition (all $|r| < 0.17$, all $p > 0.27$). The only nominal signal is a weak correlation between raw FAERS report counts and the ratio ranking ($r = 0.31$, $p = 0.035$), which may reflect prescription volume rather than toxicity. We do not read the absence of a rate-based association as evidence that selectivity and toxicity are unrelated. FAERS report counts likely track how often a drug is prescribed, and label discontinuation depends on the patient population treated for each indication, so neither can isolate the effect of selectivity. A proper test would require indication-stratified, treatment-related adverse-event rates from controlled trials with matched comparators, which are not available in standardized form across these drugs.

The candidate's distributional consistency rests on fixing its activity boundary at the assay floor x_0, and in none of the four datasets is x_0 a measured detection limit. On Davis and Klaeger it is the value assigned to pairs with no detected binding, on Metz the lowest reported value (Methods), and on Anastassiadis a 20% inhibition noise level that we set. Fixing x_0 therefore removes the analyst's choice only to the extent that each dataset's encoding of non-detection is accepted as given. Where that encoding is itself uncertain, the candidate's standing on D3 is no better than the entropy's.

Finally, D3 is assessed empirically, on four datasets. Whether the properties are jointly satisfiable only by measures within the candidate's family (Supporting Information, Section S5), and whether a uniqueness result analogous to Shannon's holds, are left for future

work. A complementary direction is a consensus ranking that does not commit to any single measure, that is, a partial order on which all existing measures agree.

The four properties are also unlikely to be the complete set, and this work suggests further candidates that we do not assess here. A measure should be correctly oriented against a crude external proxy for the concept, such as the number of active kinases, since a measure can satisfy D2 to D4 and still rank promiscuous compounds as selective. A measure should declare its scale type, because the four definitions report on incomparable scales and Figure 1 shows that leaving the operationalization undeclared can change a ranking substantially. The panel-size behaviour reported above could itself be stated as a property. Stability under measurement error in the affinities is another candidate, which the present data cannot test. Agreement across datasets profiling the same compounds can be tested, as the platform comparison above does. Klaeger and Anastassiadis share 19 compounds by name, but a comparison between an affinity assay and a single-concentration inhibition assay is harder to interpret. Metz identifies compounds only by internal ID and PubChem SID, and we did not resolve those against the other datasets.

Conclusion

We have proposed four properties that a well-formed kinase inhibitor selectivity measure should satisfy: a reliability threshold condition, bounded gap sensitivity, distributional consistency, and monotonicity under weak off-target addition. None of the four definitions assessed satisfies all four. The one property all four violate is the reliability condition, a reporting convention that any of them could adopt.

The properties come from characterizing the definitions on four datasets. Rank instability within and between definition families is associated with binding profile shape and has three structurally distinct sources. How the definitions relate depends on how they are computed: on a common log-affinity scale they split into two groups that measure different properties of

binding profiles (the S-score, entropy and Gini agree more with one another than with the ratio), but under the association-constant form of the original publication the entropy does not cluster with the S-score and Gini.

The candidate we propose, a gated, smooth-hinge effective-target number, satisfies all four properties in the following ways:

(i) the reliability condition by a reporting rule;

(ii) bounded gap sensitivity vacuously;

(iii) distributional consistency because its activity boundary is fixed at the assay floor (swept as a free parameter, that boundary moves the ranking about as much as the entropy's baseline does); and

(iv) monotonicity under weak off-target addition by its functional form, which we prove.

The candidate recovers its full-panel ranking from about a tenth to a quarter of the panel on Davis, Klaeger, and Anastassiadis, as the entropy does, and variants of it satisfy the four properties on Davis and Klaeger (Supporting Information, Section S5). It ranks the compounds almost identically to the entropy, so on these data it is not an improvement on the entropy. The case for the properties therefore does not rest on these data: they are requirements on the measure itself, and we propose them as checks that any new selectivity measure should pass.

The main goal of this work is to identify criteria for judging a candidate that could be considered a principled, universally adopted choice of selectivity measure in the future. Such criteria must be justified (which we argue they are), and jointly satisfiable (which the candidate demonstrates). Further properties may discriminate where these four do not, and the candidate may separate from the entropy on panels whose active fractions differ from the four studied here. Establishing additional properties that uniquely characterize a selectivity measure, as Shannon's axioms do for entropy, is a natural next step, for which additional datasets may be required.

The immediate practical implication is that selectivity assessments should report the sensitivity of rankings to parameter choice, and should explicitly flag compounds with no binding above the activity threshold as unreliable. The longer-term implication is that the field needs a principled basis for choosing among selectivity definitions. The need for a principled measure is not specific to selectivity. As future and ongoing work, we are developing a general measurement-theoretic framework for defining and comparing the competing measures of one concept, reporting where they agree and which pairs they leave undecided, with kinase selectivity as one instance alongside cases as varied as biological age and variant effect prediction.

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Data and Software Availability

All data and software used in this study are publicly available. The Davis kinase inhibitor profiling dataset is available at `https://github.com/dingyan20/Davis-Dataset-for-DTA-Prediction`. The Klaeger et al. dataset is available as supplementary data to the original publication

(Science, 2017, https://doi.org/10.1126/science.aan4368). The Anastassiadis et al. (*Nat. Biotechnol.* 2011) and Metz et al. (*Nat. Chem. Biol.* 2011) datasets are available as supplementary data to their original publications (https://doi.org/10.1038/nbt.2017; https://doi.org/10.1038/nchembio.530). All analysis scripts, processed data files, and the LaTeX source for this manuscript are available at https://github.com/polinavino/kinase-selectivity-definitions; this repository includes the code and full per-definition results for the secondary clinical-outcome analysis (FAERS adverse-event report rates, obtained via openFDA queries, and FDA-label discontinuation rates, from FDA product labels, for the FDA-approved Klaeger compounds). No confidential data were used in this study. All results reported in this manuscript are fully reproducible from the publicly available data and code. A preprint of this manuscript is available on ChemRxiv at https://doi.org/10.26434/chemrxiv.15001618/v1.

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Conflict of Interest

The author declares no conflict of interest.

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